



Department of Computer Science and Engineering

Mapping with Programme Outcomes and Programme Specific Outcomes

Applicable Program Outcomes (POs) with Justification

PO	Applicable Outcome and Justification
PO1: Engineering Knowledge	Applied computer science, artificial intelligence, machine learning, data processing, weather-data analysis and software engineering concepts to develop the Multi-Source AI for Hyperlocal Rainfall Nowcasting system.
PO2: Problem Analysis	Analysed the limitations of conventional weather forecasting and the challenges of localized rainfall prediction, including sparse observations, missing values and differences between weather-data sources.
PO3: Design/Development of Solutions	Designed and developed a multi-source weather intelligence system integrating IMD AWS, Personal Weather Stations, ECMWF NWP data, data processing, AI/ML nowcasting, REST APIs and a web application.
PO4: Conduct Investigations of Complex Problems	Conducted investigation and experimentation on weather observations, preprocessing, feature engineering and rainfall prediction approaches to study short-term rainfall behaviour and evaluate suitable AI/ML techniques.
PO5: Engineering Tool Usage	Used modern engineering and IT tools including Python, FastAPI, Next.js, React, Pandas, NumPy, scikit-learn, XGBoost, PyTorch, xarray, cfgrib, Matplotlib, Cartopy, SQLite and REST APIs.
PO6: The Engineer and The World	Developed a weather intelligence solution intended to improve localized weather awareness by combining real-time observations, short-term rainfall nowcasting and longer-range forecast information for the Puducherry and Tamil Nadu region.
PO7: Ethics	Applied responsible engineering practices by considering data quality, prediction uncertainty, system limitations and responsible presentation of AI-based rainfall predictions without treating forecasts as guaranteed outcomes.
PO8: Individual and Collaborative Team Work	Worked collaboratively in planning, designing, implementing and integrating the data acquisition, data processing, AI/ML, NWP, backend and web application components.
PO9: Communication	Communicated the system architecture, methodology, implementation process, weather-data workflow, AI/ML approach and findings through technical documentation, diagrams, presentations and demonstrations.
PO10: Project Management and Finance	Planned the project as modular components, organized development activities, managed implementation tasks and used suitable open-source software tools and APIs to develop the system efficiently.
PO11: Life-Long Learning	Required continuous learning of emerging technologies in AI/ML, spatiotemporal modelling, weather-data processing, numerical weather prediction, web technologies and geospatial visualization.

Applicable Program Specific Outcomes (PSOs) with Justification

PSO	Applicable Outcome and Justification
PSO1: Computational Skills	Applied fundamental and advanced computer science concepts including Python programming, data preprocessing, machine learning, REST API development and web technologies to build the hyperlocal rainfall nowcasting platform.
PSO2: Studios Research	Explored current research approaches in rainfall nowcasting and investigated Random Forest, XGBoost, CNN/LSTM and GNN-based approaches for short-term rainfall prediction.
PSO3: Placement Opportunities	Developed practical industry-relevant skills in Python, FastAPI, machine learning, data processing strengthening AI/ML readiness.